

Enhancing large language model (LLM)

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1 Background

Historically, one of the major challenges facing trustworthy language models is how to prevent them from exhibiting discriminatory behavior or unfair bias against specific groups based on protected/sensitive attributes (such as race, gender, or religion).

Therefore, this project, based on the 2020 "Metamorphic Testing and Certified Mitigation of Fairness Violations in NLP Models" [1], first tried to reproduce the process by introducing an automated software testing method called Metamorphic Testing (MT) and combining it with a stochastic smoothing framework to provide certified fairness guaranties for NLP models. Subsequent work on sample expansion and result optimization will be presented.

2 Motivation

The primary motivations for replicating and exploring this research stem from two distinct limitations in current NLP bias workflows:

- 1) **The Infeasibility of Traditional Pre-defined Rules.** Existing automated tools rely on manual, hand-coded token substitution rules (e.g., mapping "John" to "Mary"). These configurations fail to capture complex linguistic contexts and cannot scale to arbitrary unstructured sentences.
- 2) **Lack of Provable Fairness Guarantees.** While empirical debiasing techniques (e.g., fine-tuning on balanced datasets) smooth out apparent biases, they provide no mathematical lower bound on fairness, the model can still violate fairness constraints. There is an urgent need to define a mathematically sound metric, such as (ϵ, k) -fairness, and to design a mechanism capable of providing certified mitigation against fairness violations.

3 Technique

The technical core of the project consists of two phases: Automated Fairness Bug Detection (via Metamorphic Testing) and Certified Fairness Mitigation (via Randomized Smoothing).

3.1 Key Technical Steps

- 1) **Metamorphic Relation Definition:**
The framework defines an MR asserting that if an input text x is perturbed in x' by substituting sensitive attributes (without altering the core semantic context), the classification or scoring output of the model must remain statistically invariant.
- 2) **Context-Aware Mutation:**
This step utilizes a part-of-speech tagger and a masked language model to automatically swap sensitive demographic words with contextually natural alternatives, eliminating the need for manual, hand-coded templates.
- 3) **Randomized Smoothing for NLP:**
Final step is mitigating bias by injecting controlled semantic noise—such as token masking—into the input and aggregating the outputs via majority voting to establish a mathematically certified fairness guarantee.

3.2 Key Functions

- 1) `identify_sensitive_tokens(sentence)`
Utilizes Named Entity Recognition (NER) and Part-of-Speech (POS) tags to extract demographic tokens (e.g., names, nationalities, gendered pronouns).
- 2) `generate_metamorphic_inputs(sentence, target_group)`
Queries a masked language model to substitute identified sensitive tokens with context-appropriate equivalents from a

target demographic group, returning a mutated sentence x' .

- 3) `check_fairness_violation(model, x, x', threshold)`
Evaluates if $|f(x) - f(x')| > \text{threshold}$ for regression, or if $\text{argmax} f(x) \neq \text{argmax} f(x')$ for classification, flagging a fairness violation if true.
- 4) `smooth_predict(model, x, num_samples, noise_level)`
Generates N noisy variants of input x using semantic perturbations, passes them through the target NLP model, and returns the certified consensus prediction.

4 Experiment

1) Data & Preparation

The sample sentences in training dataset are entirely drawn from a large-scale movie review dataset, completely adhering to the settings of the original paper [2]. To preliminarily evaluate the performance and efficacy of the mitigation algorithm, the test suite contains 10,662 relatively concise movie reviews. Empirical results indicate that approximately one-third of these sentences are successfully "mutable" (fig. 1). For subsequent phases of this work, I plan to scale evaluation to a more complex full-length movie review dataset containing over 20,000 to 30,000 sentences.

2) Implementation

This project is deployed and executed on a high-performance computing cluster based on the HKUST Superpod architecture. Due to specific API accessibility constraints within the cluster environment, several dependencies were adapted and hosted locally.

- A local ConceptNet-mini database as knowledge graph
- The pre-trained GoogleNews-vectors-negative300.bin for word analogy and attribute mutators
- CNN classifier as the target sentiment evaluation model, Stanford CoreNLP and pytorch-human-performance-gec [3] as the same process. These dependencies are strictly aligned with the original demo.

5 Evaluation & Future Work

From the source code structure and the algorithm's sensitivity curve, quantitative results

```

=====
Input: data/dataset.txt
k value: 15
epsilon: 2.0
=====
Loading local ConceptNet database...
Local ConceptNet loaded.
stanford corenlp model loading
stanford corenlp model loaded
word2vec model loading
word2vec model loaded
| dictionary: 267744 types
# Batch mode running: data/dataset.txt

=====
total_sentences: 11289
total_mutated_sentences: 3417
total_cases: 11559
total_violations: 850
total_mitigated: 789
=====

```

Figure 1: The experimental results, with $k = 15$ and $\epsilon = 2$ on the initial sample set, generated 11559 mutation examples for 3417 variable sentences.

Table 1: Comparison of Fairness Violations Across Different ϵ Values

Parameter (ϵ)	Original Violations	Mitigated Violations
2.00	850	789
1.00	866	755
0.05	859	661

under three different parameter configurations show that the number of remaining violations is still relatively high [table 1]. Possible reasons include:

In demo code structure, the continuous prediction confidence values are normalized using a -0.5 shift. Because the target evaluation model is a CNN classifier trained on highly polarizing sentiment data (movie reviews), its outputs are heavily prone to sigmoid saturation, producing confidence scores extremely close to the boundaries (e.g., $+0.499$ or -0.499) for strong expressions. If the original sentence yields a weak positive score (e.g., $+0.02$) while a highly toxic or stereotypic mutant generated via Word2Vec or ConceptNet triggers a saturated negative response (e.g., -0.49), the absolute numerical disparity is massive. Even when ϵ scales down to 0.05, the sheer numerical magnitude of the saturated biased score easily swallows up the weak counter-weight of the original sentence.

Meanwhile, the smoothed formula adopted to calculate calibrated decision scores is expressed as:

$$\text{Score} = \frac{\sum \text{Predictions}}{e^\epsilon + k} + \frac{e^\epsilon \cdot x_{\text{test}}}{e^\epsilon + k} \quad (1)$$

Even when $\epsilon = 0.05$, where $e^{0.05} \approx 1.05$, the independent impact of x_{test} is minimized. At this

point, the sign of the final score is predominantly determined by the first term $\sum \text{Predictions}$, which denotes the sum of prediction values of the original sentence and all its generated variants. Nevertheless, if heavily gender-biased mutated sentences yield drastically divergent sentiment classification results, their aggregated sum $\sum \text{Predictions}$ will form an irreversible biased vector. This effect overwhelms the balanced weight of the original sentence, ultimately rendering the mitigation mechanism ineffective.

To address the low robustness inherent in the demo’s smoothing arch based on global mixture aggregation, I tried an optimized, pairwise simplified formulation for post-processing mitigation. The calibrated score for each individual test case is computed as follows:

$$\text{Score}_{\text{new}} = \frac{x_{\text{orig}} + e^{\epsilon} \cdot x_{\text{test}}}{1 + e^{\epsilon}} \quad (2)$$

where x_{orig} represents the sentiment score of the original seed sentence, and x_{test} denotes the score of a specific perturbed variant.

By decoupled tracking of individual test cases rather than accumulating global averages, this pointwise defense effectively isolates toxic or highly saturated outliers from contaminating the entire mutant neighborhood. Upon re-executing the evaluation pipeline with this rectified formulation, empirical results confirm that the fairness mitigation capability has been substantially optimized, manifesting a more robust control curve over fairness violations [table 2].

Table 2: Comparison of Fairness Mitigation Performance After Optimization Formulation

ϵ	Original Reduction	Optimized Reduction
2.00	7.18%	19.15%
1.00	12.82%	30.66%
0.05	23.05%	43.69%

References

- [1] Pingchuan Ma, Shuai Wang, and Jin Liu. Metamorphic testing and certified mitigation of fairness violations in nlp models. In Christian Bessiere, editor, *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 458–465. International Joint Conferences on Artificial Intelligence Organization, 7 2020. Main track.
- [2] Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Proceedings*

of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, pages 142–150, Portland, Oregon, USA, June 2011. Association for Computational Linguistics.

- [3] Tao Ge, Furu Wei, and Ming Zhou. Reaching human-level performance in automatic grammatical error correction: An empirical study, 2018.

Minutes of the 1st Project Meeting

Date	Monday morning, March 16th, 2026
Time	09:00 AM
Place	via Zoom: https://hkust.zoom.us/j/7305996251
Present	YANG Fengshuo Prof Wang Shuai
Apology	None
Note-taker	YANG Fengshuo

1. Approval of minutes

This was the first group meeting, so there were no minutes to approve.

2. Discussion items

During this meeting,

- Students Yang first report the project progress. He has thoroughly reviewed three selected academic papers and presented personal understanding of research orientations, core characteristics, adopted techniques and algorithms.
- Subsequently, the prof.Wang provided detailed guidance. He affirmed the current progress, highlighted the key points of the papers, and offered suggestions on the follow-up research direction.
- The subsequent project focuses on environment configuration and experimental reproduction based on the selected papers.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:25 AM. The next meeting will be held in April 20.

Minutes of the 2nd Project Meeting

Date	Monday morning, April 20th, 2026
Time	09:00 AM
Place	via Zoom: https://hkust.zoom.us/j/7305996251
Present	YANG Fengshuo Prof Wang Shuai
Apology	None
Note-taker	YANG Fengshuo

1. Approval of minutes

The minutes of the last meeting have been adopted.

2. Discussion items

During this meeting,

- Students Yang first report the project progress. He attempted to reproduce the experimental results from the source code but encountered several challenges, and consulted the professor regarding these difficulties..
- The prof.Wang commented on the current progress. He elaborated that the main difficulty lies in the selection of knowledge graphs, and advised the student to complete result reproduction promptly to advance further work.
- Subsequent research will be carried out in the next semester, with the primary goal of improving baseline performance.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:20 AM. The next meeting will be held in the next semester .